Brian Kang

bsg.kang@mail.utoronto.ca | Portfolio | LinkedIn | +1 647-901-9338

Education

Robotics | Engineering Science - University of Toronto Sep 2022 - Apr 2027

- **3.71 GPA** - *Dean's Honour List*

Experience

Trackside Engineering Placement - Red Bull Powertrains July 2025 - July 2026

Product Validation & Data Science Engineer

- Developed a **competitor analysis methodology**, reverse engineering **battery deployment strategies** from onboard audio via **FFT and spectrogram analysis**, identifying deployment/harvest regions to inform lap simulation and energy management strategies
- Built a **genetic algorithm optimizer** for **laptime-optimal power unit allocation** across a full race season, dynamically re-optimizing as races occur and evaluating additional PU introductions against grid penalties.
- Developed a **lumped parameter thermal model** of the 2026 MGU-K, deployed on the TAG-700 race ECU achieving **±1 °C temperature prediction** live across all 4 cars to satisfy **FIA regulatory requirements**.

Power Unit Engineer

- Operated 2026 power units on **V6 dynamometer test cells**, monitoring **ATLAS and SystemMonitor** telemetry for **full race weekend simulations** to characterize power unit performance and operational limits.
- Conducted new **engine pass-off programs** on dynos, running bearing cycles, performance validation via power curves, and **injector calibrations** to maximize power output before track usage.
- **Race operations room support** for trackside PU Engineers for Red Bull Racing and VCARB, monitoring live telemetry on ATLAS, logging faults, and tracking oil sample/wear metal data.

Thesis

Closed-Loop Control for Tendon Driven Continuum Robots with Continuous-Time State Estimation

- Continuum Robotics Laboratory, University of Toronto, Supervised by Dr. Jessica Burgner-Kahrs

Personal Projects

6-DOF Robotic Manipulator July 2026 - Present

- Designed and built a **6 DOF robot arm** with a **1 metre reach**, driven by six stepper motors, mechanical and hall-effect limit sensing for homing, and a **STM32-based motion control system**.
- Developed the complete **electromechanical system**, including 3D printed arm linkages, torque reduction, motor driver circuitry, custom PCB, power distribution, and **embedded control hardware**.
- Designed the complete **electromechanical system**, including custom mechanical arm linkages, **torque reduction**, power distribution, and motor driver circuitry.
- Built an **STM32F446RE based motion controller** with open-loop step counted position tracking, and designed a **custom serial protocol** for STM32 to host communication, bridged to **standard ROS2 topics**. Currently developing **forward/inverse kinematics** and **trajectory generation.**

Machine Learning Football Analytics Tracker July 2024 - Aug 2024

- Developed a **computer vision pipeline** using **PyTorch, OpenCV, and YOLOv5** to perform object detection and classification of players and ball, achieving **95% detection accuracy.**
- Implemented tracking and analysis algorithms to compute **team possession metrics and player speed.**

Skills

Programming: Python, C/C++, Java, MATLAB, Simulink, Docker, Kubernetes, SQL, GraphQL
ML & Robotics: PyTorch, TensorFlow, ROS2
Embedded Systems: STM32, ESP32, RISC-V, Assembly, FPGA, SystemVerilog, Quartus, ModelSim
CAD: OnShape, KiCad Solidworks, Fusion360, AutoCAD, SPICE
Hardware: 3D Printing, Laser Cutting, Oscilloscope, Soldering, Machining